Similar Triangles; Indirect Measurement

Teacher's Edition

The Period

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Must Decide similarity from all three side ratios; match the sides of a turned pair (by size, or by a name's letters); find missing sides with k = copy ÷ original; measure with a shadow, a mirror or a ramp when the words say the triangles are similar.

Must not AA named or proved (Geometry); angle sums (Thread B); a mirror or ramp item that does not say its triangles are similar.

Target I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

MTR **MTR.2.1** — Notes I: one pair as a picture, three ratios and one k · **MTR.4.1** — boards 4 and 5 split; the re-vote · **MTR.7.1** — Notes III, Example 2 and boards 6–8: heights measured without climbing

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Similar triangles have equal angles, and every side is multiplied by the same k.
2. Match by size or by angle, never by place on the page; k = copy ÷ original.
3. Same sun, same angle: shadows, mirrors and ramps make similar triangles.

Slide by Slide

1. **Similar Triangles; Indirect Measurement.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 is today's k: copy ÷ original, 4 ÷ 10 = 0.4 — a copy smaller than the original. *1 retrieves 6.06 — a perimeter, the slanted side by its triangle (√13: a whole radicand up to 225, GR.1.2 Clarification 3) (yesterday). 2 retrieves 6.04 — a ladder: the ladder is c, so subtract (last week). 3 retrieves 5.01 — k = copy ÷ original; a k under 1 shrinks (last unit). 4 retrieves MA.7.GR.1.5 — a scale drawing: every length times one number (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 8.6 units 2. 12 ft 3. k = 0.4 4. 21 ft
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Similar: one k on every side. Match by size, not by place.
6. **Notes.** The names say the matching. One matching pair gives k.
7. **Notes.** Same sun, same angle: similar triangles.
8. **Example 1 — A turned pair.** Ask which side of the second triangle matches the 5 before anyone divides: the shortest, 15 — wherever it sits.
9. **Example 1 — Worked.** Sorted: 5, 7, 9 and 15, x, y — the 5 and the 15 are both shortest.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Sorted, the copy is 12, x, 28, so x matches the 5: k = 12 ÷ 3 = 4 and x = 20 — a board with about 46.7 matched by the order in the list (3 with 28).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A mirror on the ground.** Ask which lengths match: Omar's eye height with the tree's height, and his 4 ft to the mirror with the tree's 18 ft — the words say the two right triangles are similar.
13. **Example 2 — Worked.** Ground to ground: k = 18 ÷ 4 = 4.5, so the tree is 5 × 4.5 = 22.5 ft — a board with about 1.1 used k upside down (4 ÷ 18), one with 19 added 14.
14. **Your Turn — Same steps, your numbers. Boards up when done.** k = 12 ÷ 1.5 = 8, so the flagpole is 4 × 8 = 32 ft — a board with 0.5 used k upside down (1.5 ÷ 12).
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. Are triangles with sides 8, 12, 14 and 12, 18, 21 similar? → ***Yes — every ratio is 1.5*** *'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]; 'yes' from two ratios only [Notes I: all three]*

18–19. Board 2. Triangle ABC is similar to triangle DEF. Find x. → ***x = 24*** *13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 20 — added 12 to the 8 [Notes I: similar means multiplied by one k]*

20–21. Board 3. A triangle has sides of 12, 16 and 20 inches. A similar triangle's longest side is 5 inches. Find its other two sides. → ***3 inches and 4 inches*** *48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]*

22–23. Board 4. The two triangles are similar (the second is turned). Which side of the second triangle matches the 9? → ***C — the 12*** *(A) the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]; (B) 16, the longest, for the middle side [Notes I: match by size]; (D) looked for a side equal to the 9 — similar taken as equal [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. Which triangle is similar to one with sides 5, 6 and 8? → ***B — 10, 12, 16*** *(A) added 2 to every side [Notes I: similar means multiplied by one k, not added]; (C) checked two ratios, not three: 15 ÷ 8 is not 2 [Notes I: all three ratios]; (D) kept two sides the same [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow. How tall is the tree? → ***24 ft*** *about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 16.5 — added 12.5 [similar means multiplied]*

28–29. Board 7. A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar. How high is the ramp 8 m from A? → ***2.4 m*** *1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]; 6.6 — added 6 [similar means multiplied]*

30–31. Board 8. Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror. The two right triangles are similar. How tall is the flagpole? → ***20 ft*** *The flagpole's age (45 years) is not needed. 180 or 225 — used the 45 years [the extra number]; 1.25 — divided the other way, 4 × 5 ÷ 16 [Florida B1G-M, GR.2.4: mismatching corresponding sides]*

32–33. Board 9. Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2. Is she right? Explain. → ***No — similar means × one k, not + 2.*** *'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–3: similar or not by three ratios, a named turned pair's missing side (k = 3), then a k under 1; 4 votes on matching in a turned pair, 5 on which triangle is similar. The pattern breaks at 6 — a shadow — then a ramp's two triangles from their shared corner (7) and a mirror with an extra number (8); 9 is written. The independent set: similar or not (it is not), named triangles with no picture (the letters say the matching), a k under 1, a shadow, a nested pair from the shared corner (with its picture), which is similar.

Misconceptions to Watch

Board 1. 'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]

Board 2. 13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]

Board 3. 48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]

Board 4. the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]

Board 5. added 2 to every side [Notes I: similar means multiplied by one k, not added]

Board 6. about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]

Board 7. 1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]

Board 8. 180 or 225 — used the 45 years [the extra number]

Board 9. 'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]

Materials Whiteboards and markers. Calculators allowed, printed nowhere. The Unit Review, one per student, goes home today.